THREE EQUIVALENT FORMULATIONS OF A LOCAL PROPERTY OF MORPHISMS OF COMPLETE WEAKLY ADMISSIBLE RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays two large commutative diagrams and one small one; all lie inside the counted span and are reproduced verbatim.

Editorial note. The axioms are referred to by label; those references lie outside the counted span and are carried as navigation stubs rather than reproduced, and the situation governing them is recorded as a standard prerequisite.

Editorial note. The proof contains the author's parenthetical noting that applying the second axiom with identities is permissible because identities are faithfully flat ring maps; it is reproduced verbatim.

Editorial note. The counted spans carry 8 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The locality axioms for a local property of morphisms of complete weakly admissible rings. Same chapter, the situation governing these axioms, referred to at proof lines 5155-5156, 5160-5166 and 5171-5175, with the individual items item-axiom-1, item-axiom-2 and item-axiom-3. Supply the three moves by which the property is transported around the diagram.
- The definition of a formal algebraic space through coverings, and fibre products of formal algebraic spaces. Same chapter, Definition definition-formal-algebraic-space used throughout at proof lines 5119-5126, and Lemma lemma-fibre-products-general at proof lines 5133-5135. Provide the coverings whose members are affine and the statement that the fibre products appearing in the chase are formal algebraic spaces.
- Morphisms between affine formal algebraic spaces coming from completions of etale ring maps, and surjective such morphisms giving faithfully flat etale ring maps. Same chapter, Lemma lemma-etale at proof lines 5150-5153 and Lemma lemma-etale-surjective at proof lines 5177-5179. Identify the ring maps to which the axioms are applied and recognise the final covering as a faithfully flat etale extension.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0AND: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5086–5086. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 0ANE: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5090–5090. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original reference 0ANF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 5095–5095. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN8: sources/stacks-algebra/pinned-original-source.tex, source lines 39507–39507. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN9: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4492–4492. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AJ2: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3488–3488. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is a *formal algebraic space* if there exist a family of maps $\{X_i \to X\}_{i \in I}$ of sheaves such that

(1) X_i is an affine formal algebraic space,
(2) $X_i \to X$ is representable by algebraic spaces and étale,
(3) $\coprod X_i \to X$ is surjective as a map of sheaves

and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally countably indexed formal algebraic spaces over S. Let P be a local property of morphisms of $WAdm^{count}$. The following are equivalent*

(1) *for every commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with U and V affine formal algebraic spaces, $U \to X$ and $V \to Y$ representable by algebraic spaces and étale, the morphism $U \to V$ corresponds to a morphism of $WAdm^{count}$ with property P,

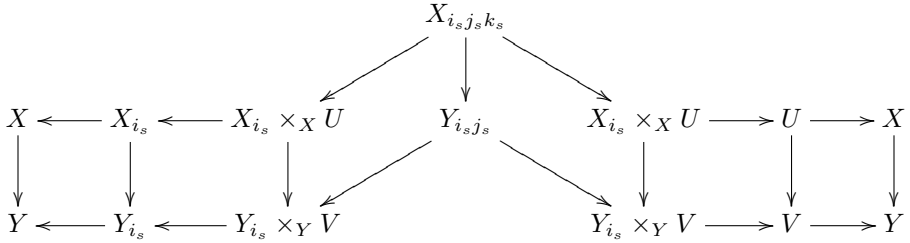
(2) *there exists a covering $\{Y_j \to Y\}$ as in Definition 0.1 and for each j a covering $\{X_{ji} \to Y_j \times_Y X\}$ as in Definition 0.1 such that each $X_{ji} \to Y_j$ corresponds to a morphism of $WAdm^{count}$ with property P, and*

(3) *there exist a covering $\{X_i \to X\}$ as in Definition 0.1 and for each i a factorization $X_i \to Y_i \to Y$ where Y_i is an affine formal algebraic space, $Y_i \to Y$ is representable by algebraic spaces and étale, and $X_i \to Y_i$ corresponds to a morphism of $WAdm^{count}$ with property P.*

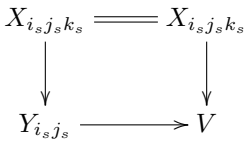
Proof. It is clear that (1) implies (2) and that (2) implies (3). Assume $\{X_i \to X\}$ and $X_i \to Y_i \to Y$ as in (3) and let a diagram as in (1) be given. Since $Y_i \times_Y V$ is a formal algebraic space (Lemma 0AJ2) we may pick coverings $\{Y_{ij} \to Y_i \times_Y V\}$ as in Definition 0.1. For each (i, j) we may similarly choose coverings $\{X_{ijk} \to Y_{ij} \times_{Y_i} X_i \times_X U\}$ as in Definition 0.1. Since U is quasi-compact we can choose $(i_1, j_1, k_1), \ldots, (i_n, j_n, k_n)$ such that

$$X_{i_1 j_1 k_1} \amalg \ldots \amalg X_{i_n j_n k_n} \longrightarrow U$$

is surjective. For $s = 1, \ldots, n$ consider the commutative diagram



Let us say that P holds for a morphism of countably indexed affine formal algebraic spaces if it holds for the corresponding morphism of $WAdm^{count}$. Observe that the maps $X_{i_s j_s k_s} \to X_{i_s}$, $Y_{i_s j_s} \to Y_{i_s}$ are given by completions of étale ring maps, see Lemma 0AN8. Hence we see that $P(X_{i_s} \to Y_{i_s})$ implies $P(X_{i_s j_s k_s} \to Y_{i_s j_s})$ by axiom (0AND). Observe that the maps $Y_{i_s j_s} \to V$ are given by completions of étale rings maps (same lemma as before). By axiom (0ANE) applied to the diagram



(this is permissible as identities are faithfully flat ring maps) we conclude that $P(X_{i_s j_s k_s} \to V)$ holds. By axiom (0ANF) we find that $P(\coprod_{s=1,\ldots,n} X_{i_s j_s k_s} \to V)$ holds. Since the morphism $\coprod X_{i_s j_s k_s} \to U$ is surjective by construction, the corresponding morphism of $WAdm^{count}$ is the completion of a faithfully flat étale ring map, see Lemma 0AN9. One more application of axiom (0ANE) (with $B' = B$) implies that $P(U \to V)$ is true as desired. □